

Has Futures Markets Financialization Changed Monetary Policy Transmission to Commodity Prices?

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Abstract

This research examines whether the financialization of commodity futures markets that happened in the first 10 years of the 2000s has changed the transmission of monetary policy to commodity prices in the US. Using an instrumented local projections strategy with high-frequency monetary policy surprises, I estimate the evolution of the causal effect of monetary policy on commodity futures prices allowing this effect to vary with two measures of financialization: the non-commercial gross share of open interest and the rolling correlation with S&P 500 returns. I find that financialization amplified monetary policy transmission for industrial commodities - copper and oil - through a risk appetite channel, while results for agricultural commodities are more muted. A subsample comparison confirms that monetary policy transmission changed significantly in the post-financialization period relative to the pre-financialization period for the former.

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1 Introduction

Futures markets for commodities have existed, in some cases, since the nineteenth century. Originally, they were traded exclusively by market operators – commercial traders – to hedge against price oscillations risks.

However, in the first 10 years of the 2000s, futures markets of commodities have seen a large rise in the share of non-commercial traders ([Cheng and Xiong, 2013](#); [Tang and Xiong, 2012](#); [Kang et al., 2023](#); [Irwin and Sanders, 2012](#)). In other words, trading commodity futures has become part of the investment strategy of many players in financial markets with no direct involvement in these commodities production and sales.

The scope of this analysis is to examine whether financialization has changed the way monetary policy affects commodity prices. There are multiple reasons to care about this.

First, commodity prices matter for households - food and energy - and for manufacturing firms - as input costs -, which makes understanding their sensitivity to monetary policy transmission relevant for policymakers. In particular, recent shocks to global supply chains highlighting how globalization has exposed the global economy to commodity price fluctuations.

Then, following two decades of absence, inflation seems to be back as a pressing macroeconomic concern. If financialization has strengthened monetary policy transmission to commodity prices, tightening cycles designed to tackle it may have larger spillover effects onto commodity markets than historical estimates based on pre-financialization data would suggest. Understanding how and why monetary policy transmission to the economy has evolved can help tailor policy responses to the new context.

Finally, the existing literature on commodity futures markets financialization has been mostly focused on its effects on price co-movement with the stock market ([Tang and Xiong, 2012](#); [Kang et al., 2023](#)) or price volatility ([Brunetti et al., 2016](#)). This research aims at adding a piece to the puzzle by exploring the effects on these markets permeability to monetary policy transmission.

We build on the well-established finding that monetary policy affects asset prices in large part

through its effects on risk appetite ([Bauer et al., 2023](#)): monetary policy tightening decreases risk-prone behavior by worsening the perceived economic and financial environment, downgrading the economic outlook, augmenting uncertainty, and weakening the balance sheets of borrowers and lenders, while easing has the opposite effect.

Before financialization, this risk appetite channel had no direct pathway into commodity markets, which were dominated by commercial hedgers whose positions reflected physical business needs rather than portfolio considerations. Financialization opened this pathway: as [Basak and Pavlova \(2016\)](#) show, the entry of institutional investors into commodity futures markets means that macroeconomic shocks are now transmitted to commodity prices through the portfolio rebalancing decisions of non-commercial investors. When monetary policy tightens, lower risk appetite leads institutional investors to decrease their commodity futures allocations as part of a broader shift away from risky assets, putting downward pressure on futures and — through arbitrage — spot prices¹. When monetary policy eases, the reverse rebalancing boosts commodity prices through the same mechanism.

To explore this mechanism I use the monetary policy shocks identified by [Acosta et al. \(2025\)](#) to build an interacted instrumented local projections strategy on US futures markets data to discover the role of financialization over the last 30 years.

The reason to focus on the US is two-fold: First, given the role of the dollar in international trade, US exchanges host the global benchmark futures contracts for all the commodities included in this study. Crude oil on NYMEX, copper on COMEX, soybeans and wheat on CBOT are the reference prices for global physical trade in these commodities, with clear consequences on global spillovers of US prices.

Second, the American financial market is as deep, sophisticated and innovative as financial markets can be, meaning that if there is an effect of financialization, it is likely to be found there.

The same analysis is run on different commodities separately to let heterogeneity in financializa-

¹Commodity spot prices represent the current market value for immediate delivery and payment, reflecting instant supply and demand

tion patterns across different types of commodities - industrial and agricultural - to emerge. I also test two different measures of financialization to find which particular aspect of this phenomenon is more relevant.

1.1 Literature Review

Various papers have documented the surge in financial investment in commodity futures markets. [Tang and Xiong \(2012\)](#) and [Kang et al. \(2023\)](#) document an increase in non-commercial traders presence in these markets across different commodities from the start of the 2000s. Their presence also resulted in these futures becoming correlated with S&P500 equity returns, signaling the effect of financial investors that treated these futures as risky asset investments. [Cheng and Xiong \(2014\)](#) find a structural break around 2004, with the presence of non-commercial traders increase accelerating in the years previous to the Financial Crisis. With the caveat that the effects of this phenomenon have relevant variation across different commodities, [Domanski and Heath \(2007\)](#) find that financial investors began to deal with commodity futures as any other financial trade. Finally, [Irwin and Sanders \(2012\)](#) showed how agricultural commodities saw a partial de-financialization after 2014, a result that will be useful in analyzing this paper's results.

In the last decade, a debate unfolded on the effectiveness monetary policy in influencing commodity prices. [Scrimgeour \(2015\)](#) uses a high frequency approach to show that monetary contractions have a significant negative effect on commodity prices, which varies significantly between industrial - stronger - and agricultural commodities - weaker -, with noisy results for oil. [Hammoudeh et al. \(2015\)](#)'s structural VAR finds both spot and futures prices show an initial spike followed by a persistent decline - especially for metals and other industrial commodities - when the central bank increases rates.

On the other hand, the literature on financialization's influence on monetary policy transmission to commodity prices is limited. Very recently, [Ivan et al. \(2025\)](#) used a high-frequency instrumental variable estimator to find that unexpected monetary policy contractions reduce the trading volume

of commodity futures. Monetary policy tightening reduces commodity futures liquidity, and this effect is strongest — and most price-relevant — for the most actively traded contracts. They identify a liquidity channel: higher rates decrease liquidity availability for financial investors, who then reduce their exposure to these markets and put downward pressure on prices.

This paper concurs to address the gap in the literature and complements their findings in three ways. First, it examines how the strength of monetary policy transmission has evolved over time with financialization, providing a historical perspective that high-frequency event studies cannot capture. Second, it traces the medium- and long-run cumulative price response over a 24-month horizon rather than the immediate announcement-day impact. Third, it identifies the risk appetite channel by measuring financialization through trader composition and equity co-movement rather than trading volume, capturing a distinct dimension of financial integration.

The rest of this dissertation is organized as follows. [Section 2](#) and [Section 3](#) present my data and main empirical strategy respectively, while results are illustrated and discussed in [Section 4](#). Finally, [Section 5](#) outlines my conclusive remarks and possible future research agenda.

2 Data

2.1 Financialization measurement

I use two measures of financialization to capture two different aspects of this phenomenon.

First, I consider the non-commercial gross share of open interest (NC gross share)², as calculated by the US Commodity Futures Trading Commission³. It is the most widely used measure in the literature and it counts how many futures contracts are held by non-commercial operators relative to the total number of existing contracts. By non-commercial traders, they mean large traders not

²The "gross" share sums long and short non-commercial positions divided by twice total open interest, rather than netting them. This captures total financial investor participation regardless of directional positioning.

³The COT reports are based on position data reported by firms. However, the actual trader category or classification is based on the self-reported business purpose and is subject to review by CFTC staff. For details, see <https://www.cftc.gov/MarketReports/CommitmentsofTraders/index.htm>

using futures for hedging, such as hedge funds and managed money.

The scope is to describe the penetration of financial investors in commodity futures markets - extensive margin. Time series monthly - end of month - data were retrieved from 1994 to 2026 for each commodity ([Commodity Futures Trading Commission, 2024](#)). I compute a 12-months rolling average to smooth the process.

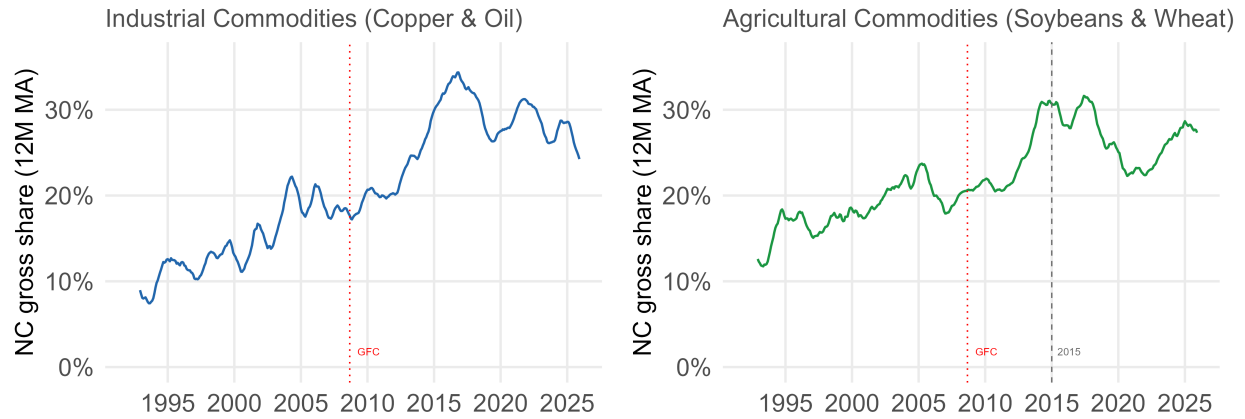
The other measure is a 24-month rolling correlation of commodity futures prices with the S&P 500 index, which is less common in the existing literature. Indeed, the flow of investments into commodity futures markets was primarily of two types: index investment and financial speculation. The former is when an investor buys a set of indexed assets: they are not interested in investing in commodities specifically, they are just diversifying their portfolio. Thus, they are less likely to be influenced by the risk appetite channel. The latter, instead, is a more flexible type of investment, which is aimed to make profits through risk assets trading: thus, they are driven by risk appetite and subject to monetary policy influence. Therefore, including the co-movement with equity returns tells a difference but crucial nuance of the story. I obtained the data from the [Federal Reserve Bank of St. Louis \(2024\)](#) and computed the index with commodity futures prices data. This measure should tell how deeply the presence of financial investors changed the nature of commodity futures markets by making them behave more like conventional risky financial assets - thus, with a positive co-movement with equity prices. It represents the intensive margin of the analysis.

The two indexes combined capture most of the financialization phenomenon.

2.2 Data overview and sample construction

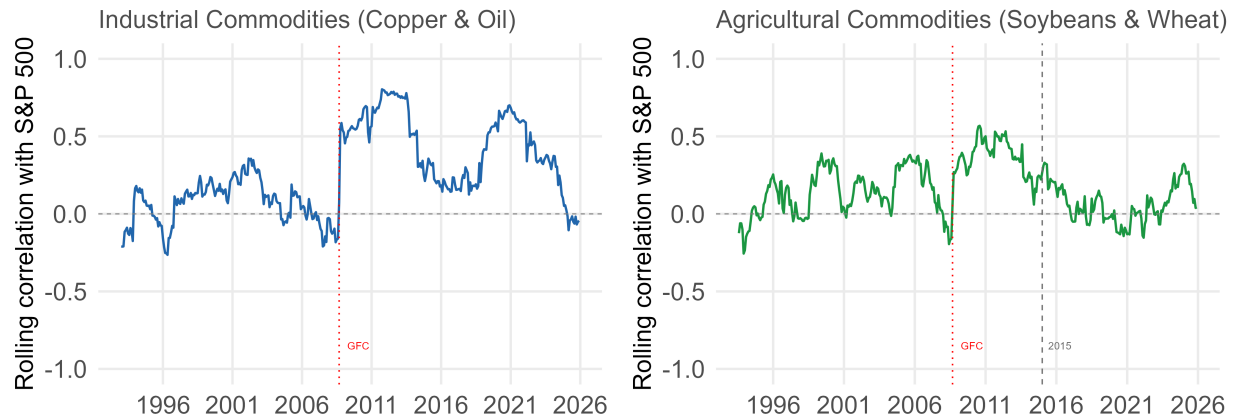
I have selected a sample of 5 commodities to analyze: crude oil, copper, wheat, soybeans, and gold. The aim is to cover different types of commodities across different sectors – industrial inputs and agricultural products – and different financialization behavior.

Figure 1: Non-Commercial Gross Share of Open Interest



Note: Monthly non-commercial gross share of open interest, smoothed with a 12-month rolling average. Equal weights within each group. Red dotted line: Global Financial Crisis (September 2008). Grey dashed line: onset of agricultural de-financialization (2015). Sample: 1990–2026. Although raw data are available from 1990, the 12-month rolling average means the smoothed series begins in January 1990, as the first twelve observations are required to compute the initial value.

Figure 2: 24-Month Rolling Correlation with S&P 500



Note: 24-month rolling correlation between monthly log returns of commodity futures and the S&P 500 index. Equal weights within each group. Red dotted line: Global Financial Crisis (September 2008). Grey dashed line: onset of agricultural de-financialization (2015). Sample: 1990–2026. Although raw data are available from 1990, the 12-month rolling average means the smoothed series begins in January 1990, as the first twelve observations are required to compute the initial value.

Figure 1 and Figure 2 represent the time series financialization data for each of the two measures respectively comparing industrial - copper and oil - and agricultural commodities - wheat and soybeans. Overall, the two indexes considered move consistently over the period considered – January 1990 - January 2026.

Comparing industrial and agricultural commodities confirms our prior that financialization was

sharper and faster for industrial commodities; agricultural commodities saw an increment in both indexes as well until 2015, and then experienced the partial de-financialization identified by [Bianchi et al. \(2020\)](#).

The NC gross share of open interest shows a clear rising trend that peaks around 2015 for both commodity groups and then stabilizes for oil and copper, while it starts declining for wheat and soybeans. On the contrary, the correlation with S&P 500 proceeds gradually upwards for agricultural commodities and then declines sharply after 2015; industrial commodities show a clear structural break around the financial crisis, with the average post-crisis level of correlation around roughly doubled with respect to the pre-crisis period. The slight difference in size of de-financialization of wheat and soybeans encountered in the two indexes indicates that the non-commercial traders' penetration in these futures markets did not decrease much, but seems to suggest that their behavior changed causing a decline in the correlation with the S&P index.

Interestingly, the Great Financial Crisis seems to have revealed the financial transformation of commodity markets. In fact, both commodity groups experienced a sharp positive increase in the co-movement with the financial markets: this did not happen because new investors arrived - the NC gross share shows no abrupt departure from trend -, but because the investors already present demonstrated for the first time that they would treat commodities as risk assets in a crisis. Despite smoothing, both measures keep quite relevant noise, particularly in the case of correlation with stock prices. This is not surprising, given that prices tend to oscillate significantly in financial markets, while portfolio re-balancing, which affects the movements in non-commercial share of open interest requires more time to adjust. [Section A.2](#) presents breakdowns for single commodities financialization measures. The within-group comparison does not show significant divergence between oil and copper and wheat and soybeans.

Finally, I decided to analyze gold because of its peculiarity. Gold is a commodity only in name, since the relative lower relevance of industrial use makes it closer to a pure financial asset; nevertheless, it is a commodity with some storage costs - even if low relative to its value because it is dense, it does not corrode, degrade, or evaporate, and requires only secure vault space. Finally, it is

also a safe haven in times of crisis. All these aspects contribute to make the analysis of gold more complicated, but also allow to use gold to add useful insights on our analysis. For example, the almost complete absence of the storage channel permits to check its presence in other commodities' analysis.

Note that this nuances the meaningfulness of both financialization measures for gold⁴. The non-commercial share is problematic because banks are considered as commercial traders, while the S&P500 correlation because gold is a form of investment when the economy is booming but also a safe haven in economic crises. However, the idea is that if monetary policy effects were roughly stable for gold and gold represented the upper bound of this effect quantitatively, the financialization channel theory is supported.

Commodity spot and futures price data are obtained from Refinitiv (London Stock Exchange Group, LSEG) Datastream ([London Stock Exchange Group \(LSEG\), 2024](#)). Daily settlement prices are collected over the sample period 01.01.1990 - 01.01.2026. For each commodity, the front-month futures contract is used, as detailed in [Table 1](#). The last available daily settlement price of each month is retained to construct a monthly series. Monthly log returns are then computed as:

$$p_{i,t} = \ln(P_{i,t}) - \ln(P_{i,t-1}) \quad (1)$$

where $P_{i,t}$ denotes the end-of-month settlement price of the front-month futures contract for commodity i .

For US monetary policy - my main regressor -, I use monetary policy shocks from [Acosta et al. \(2025\)](#) drawn from the U.S. Monetary Policy Event-Study Database ([Federal Reserve Bank of San Francisco, 2026](#)) for the exogenous component needed for an IV-LP strategy. The shock is constructed as the first principal component of intraday changes in money market futures rates over approximately a one-year horizon, measured over windows covering both the FOMC statement re-

⁴[Section A.2](#) portrays the two indexes times series for gold as well.

lease and the Chair’s press conference where applicable. I transformed raw event-level surprises in units of a standard 25 basis point rate change and aggregated to monthly frequency by summing all surprises occurring within each calendar month. Months containing no FOMC event are assigned a value of zero. Note that given that the shock series is available only from January 1994 onward, the final sample has been restricted accordingly.

As to the endogenous component, I used the monthly change - first difference - in the one-year Treasury constant maturity yield. Daily data are retrieved from the Federal Reserve Bank of St. Louis FRED database (series DGS1), published by the [Board of Governors of the Federal Reserve System \(US\) \(2026\)](#) and expressed in percent per annum. The last available daily observation of each month is retained to construct a monthly end-of-month series, consistent with the price series construction described above. I use this series rather than policy rates or Fed Funds Futures because it captures, because I wanted to include the aspects of monetary policy not strictly related to setting the policy rate - e.g., forward guidance.

3 Empirical Strategy

This section describes the empirical strategy I employed for this analysis. In [Section 3.1](#) I present my empirical framework. [Section 3.2](#) shows the baseline model that computes monetary policy impact on these commodity futures prices over the whole sample. This serves as a benchmark. Then, [Section 3.3](#) presents the interaction model that investigates whether financialization amplifies monetary policy, while [Section 3.4](#) shows the subsample analysis that compares the pre-financialization and post-financialization monetary policy effects.

3.1 Local Projection IV Framework

To estimate how financialization changes the transmission of monetary policy to commodity futures prices, I employ the Local Projections estimator of [Jordà \(2005\)](#), augmented with an instrumental variable identification strategy. The resulting LP-IV estimator recovers horizon-by-horizon

impulse response functions (IRFs) that trace the cumulative log price response of each commodity to an exogenous monetary policy shock over a 24-month horizon. The choice of a 24-month horizon balances the need to capture the full portfolio rebalancing dynamics of financial investors — which are not expected to materialise immediately — against the loss of observations entailed by forward-shifting the dependent variable. Given a sample of 384 months, the 24-horizon specification retains a sufficient number of observations at each horizon while covering the medium-run transmission dynamics of interest.

The model is estimated separately for each commodity, which serves two purposes. First, it allows the IRF to be fully commodity-specific, capturing the heterogeneous transmission mechanisms documented in the literature ([Scrimgeour, 2015](#); [Hammoudeh et al., 2015](#)). Second, it enables the inclusion of commodity-specific controls, including producer-country exchange rate returns and commodity-specific crisis dummies, which would be absorbed by fixed effects in a pooled specification but would otherwise contaminate the identification of the monetary policy channel.

The instrument of [Acosta et al. \(2025\)](#) is preferred over the widely used alternative of [Gertler and Karadi \(2015\)](#) for two reasons. First, Gertler and Karadi rely on Fed Funds Futures surprises as their baseline instrument, which are less informative at the zero lower bound, a period that coincides with the period of higher financialization. The [Acosta et al. \(2025\)](#) series covers a broader set of money market futures over approximately a one-year horizon, retaining meaningful variation even when the short rate is constrained. Second, the USMPD covers monetary events that include both the FOMC statement release and the Chair’s press conference, ensuring that communication surprises, which have become an increasingly important tool of monetary policy precisely in the last 15 years, are fully captured.

Finally, a potential concern with high-frequency monetary policy instruments is the information effect identified by [Nakamura and Steinsson \(2018\)](#). To assess this, I test whether the [Acosta et al. \(2025\)](#) shock is correlated with contemporaneous S&P 500 returns, that is the standard diagnostic for the information effect according to [Jarociński and Karadi \(2020\)](#). No significant correlation

is found, suggesting the instrument captures genuine policy surprises rather than Fed information releases⁵.

3.2 Baseline Specification

Before turning to the actual models, I estimate a baseline specification that recovers the price response to monetary policy tightening, abstracting from the role of financialization. The baseline serves two purposes: it connects the analysis to the prior literature on monetary policy and commodity prices, and it provides a reference point against which the interaction results can be interpreted.

In the first stage regression, the endogenous variable is the monthly change in the one-year Treasury yield, Δi_t :

$$\Delta i_t = \alpha + \beta \varepsilon_t^m + u_t \quad (2)$$

where ε_t^m is the high-frequency monetary policy surprise from [Acosta et al. \(2025\)](#), which serves as the excluded instrument. No additional controls enter the first stage because Δi_t is a macro-level variable common across commodities.

The baseline second stage is estimated separately for each commodity i :

$$p_{i,t+h} - p_{i,t-1} = \alpha^h + \delta^h \widehat{\Delta i}_t + \Psi^h \mathbf{X}_{i,t} + \varepsilon_{i,t}^h \quad (3)$$

where $p_{i,t+h} - p_{i,t-1}$ is the cumulative log price change from $t - 1$ to $t + h$, $\widehat{\Delta i}_t$ are the first-stage fitted values from equation (2), and $\mathbf{X}_{i,t}$ is a vector of commodity-specific controls. The coefficient δ^h traces the IRF at each horizon h and is the object of interest⁶

The control vector $\mathbf{X}_{i,t}$ comprises four lags of the log price change $\Delta p_{i,t}$, four lags each of industrial production growth and CPI inflation, and — where applicable — four lags of the rel-

⁵see [Figure A.11](#)

⁶Standard errors are estimated using the Newey-West HAC estimator.

evant producer-country exchange rate return.⁷ Crisis dummies for the Global Financial Crisis (September 2008 – June 2009) and the COVID-19 pandemic (March 2020 – June 2021) enter contemporaneously and with four lags; the Ukraine war dummy (February 2022 – December 2022) is additionally included for Oil, soybeans, and Wheat. For Oil only, the specification additionally includes contemporaneous and four lags of the [Känzig \(2021\)](#) OPEC supply news shock, to absorb oil supply shock contamination of the monetary policy transmission channel.

3.3 Interaction Model

The main specification augments the baseline with an interaction term between the instrumented policy shock and a lagged, standardised measure of financialization, allowing the price response to vary with the degree of financial investor participation in each commodity market. In particular, we should see the interacted model to deliver more negative (positive) coefficients in presence of a monetary policy tightening (easing): if this is the case, it means financialization moves commodity prices closer to normal asset prices. I used a commodity-specific standardization to ensure that the interaction coefficient ϕ^h is interpretable within each commodity as the marginal effect of a one-standard-deviation increase in financialization on monetary policy transmission, expressed relative to each commodity's own historical financialization distribution. The standardised financialization measure for commodity i at time t is defined as:

$$\tilde{F}_{i,t-1} = \frac{F_{i,t-1} - \bar{F}_i}{s_i} \quad (4)$$

where $F_{i,t-1}$ is the lagged financialization measure, $\bar{F}_i$ is its full-sample mean, and s_i is its full-sample standard deviation, both computed within each commodity's own full estimation sample. Since the standard deviations used for normalization differ across commodities in absolute terms, degrees of financialization amplification cannot be compared across commodity but only within the same commodity across different financialization measures and subsamples. Nevertheless, we

⁷The exchange rate controls are the Brazilian Real (`d_br1`) for Coffee and soybeans, the Chilean Peso (`d_clp`) for copper, and the Australian Dollar (`d_aud`) for Gold. No exchange rate control is included for Oil or Wheat.

can comment on coefficient significance and direction heterogeneity across commodities. This allows us to assess both whether financialization amplifies monetary policy transmission for a given commodity, and whether the two measures — the NC gross share and the rolling S&P 500 correlation — deliver a consistent picture of the amplification mechanism.

The interaction term entering the second stage is then constructed as:

$$\widehat{\Delta i}_t \times \tilde{F}_{i,t-1} \quad (5)$$

that is, the product of the first-stage fitted values with the standardised lagged financialization measure. Because $\tilde{F}_{i,t-1}$ is predetermined at time t , this product inherits the exogeneity of $\widehat{\Delta i}_t$ and serves as a valid instrument for the endogenous interaction $\Delta i_t \times \tilde{F}_{i,t-1}$.

The second stage is estimated separately for each commodity i , each financialization measure F and it modifies the baseline model as follows:

$$p_{i,t+h} - p_{i,t-1} = \alpha^h + \delta^h \widehat{\Delta i}_t + \phi^h \left(\widehat{\Delta i}_t \times \tilde{F}_{i,t-1} \right) + \lambda^h \tilde{F}_{i,t-1} + \Psi^h \mathbf{X}_{i,t} + \varepsilon_{i,t}^h \quad (6)$$

where $\widehat{\Delta i}_t$ are the fitted values from equation (2), replacing the endogenous policy rate change; $\widehat{\Delta i}_t \times \tilde{F}_{i,t-1}$ is the constructed interaction term from equation (5), replacing the endogenous product $\Delta i_t \times \tilde{F}_{i,t-1}$; $\tilde{F}_{i,t-1}$ enters directly to control for the main effect of financialization on price dynamics, ensuring that ϕ^h identifies the marginal amplification effect of financialization on monetary transmission rather than a level difference in prices across financialization regimes.

The coefficient of interest is ϕ^h , which measures the marginal effect of a one-standard-deviation increase in financialization on the log price response at horizon h . A negative ϕ^h indicates that higher financialization amplifies the negative price response to a contractionary monetary shock, consistent with our hypothesis. The specification is estimated twice per commodity, once for each financialization measure.

3.4 Subsample Analysis

To assess whether the transmission of monetary policy to commodity prices changed structurally following the financialization of commodity markets, I compare IRFs estimated on two non-overlapping subsamples: a pre-financialization period (January 1994 – December 2007) and a post-financialization period (January 2010 – January 2026). The years 2008–2009 are excluded from both subsamples to avoid contaminating the comparison with the Global Financial Crisis, which represents a confounding structural break in both monetary policy and commodity market conditions.

The subsample split is motivated by the descriptive evidence in [Section 2.2](#) and the literature summarized in [Section 1](#), which documents a sustained increase in both financialization measures beginning around 2004–2008 and stabilizing at a higher level thereafter.

The baseline specification (3) is estimated separately on each subsample, with the first stage also run independently for each sub-period to account for the possibility that the relationship between the instrument and the policy rate changed across regimes. Specifically, the first stage for each subsample takes the form:

$$\Delta i_t = \alpha_s + \beta_s \varepsilon_t^m + u_t, \quad s \in \{\text{pre}, \text{post}\} \quad (7)$$

where subscript s denotes subsample-specific coefficients. The fitted values $\widehat{\Delta i}_t^s$ are then merged into the commodity-level panel and used as the endogenous regressor in the subsample second stage.

In the second stage, the subsample comparison is estimated using a parsimonious control set relative to the baseline and interaction specifications. Specifically, currency controls are excluded and crisis dummy lags are reduced from four to two. The parsimonious specification reduces the regressor count to approximately 20 parameters, preventing the confidence bands from becoming too large, as opposed to the full specification with currency controls and four crisis dummy lags which would require approximately 30 parameters per subsample at horizon $h = 24$.

Moreover, plotting the two IRFs does not allow to claim that there was a structural change if confidence intervals do not overlap because the underlying data - futures prices - is likely to be highly correlated across the two subsamples. Instead, to assess the statistical significance of the difference in IRFs between the two subsamples, I employ a time-cluster bootstrap following the strategy illustrated in [Berthold et al. \(2026\)](#). The bootstrap resamples dates with replacement separately for each subsample, preserving cross-sectional dependence across commodities at each point in time⁸.

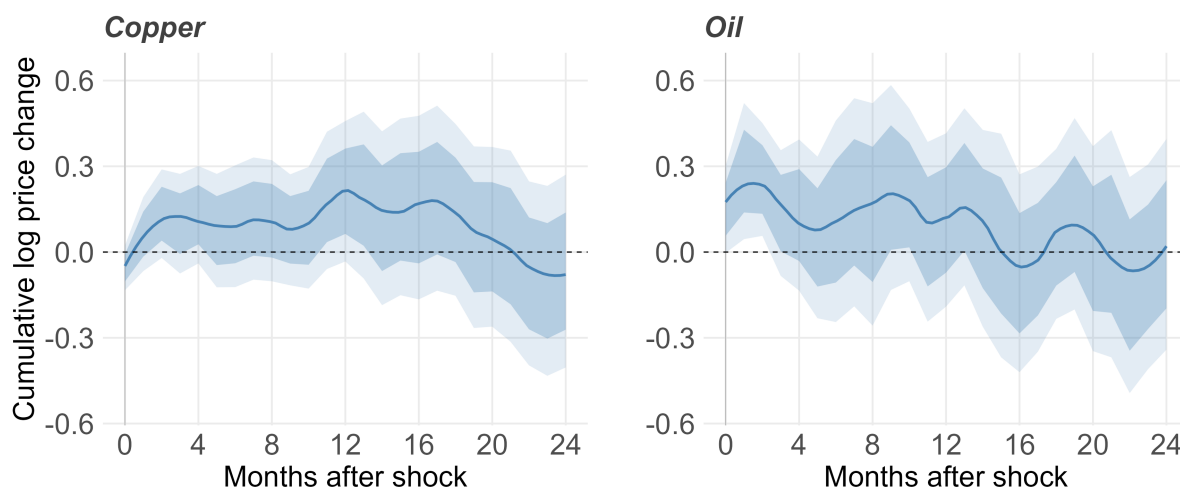
⁸See [Section A.5](#) for details on the degrees of freedom constraints motivating the parsimonious model and the full illustration of the bootstrapping procedure and its rationale

4 Results interpretation

This section is organized as follows: [Section 4.1](#), [Section 4.2](#) and [Section 4.3](#) present the results from my three main models, while [Section 4.4](#) discusses their implications.

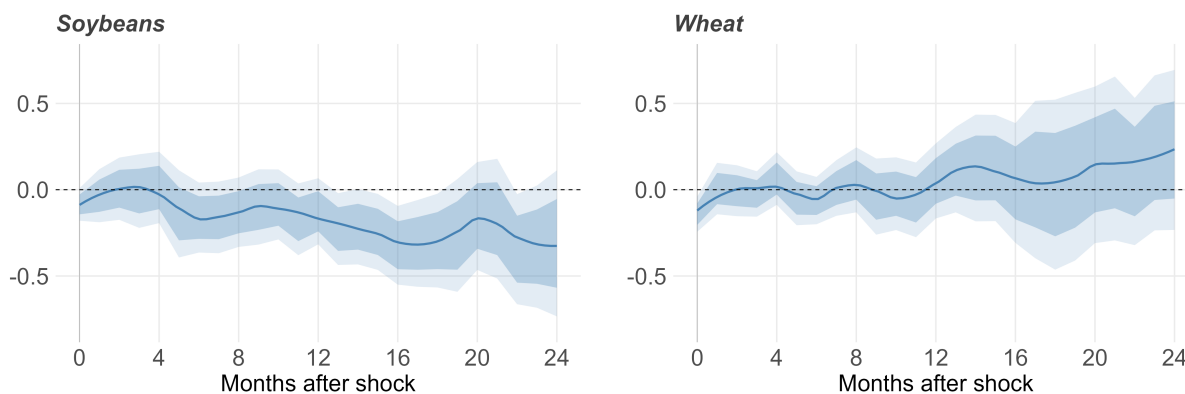
4.1 Baseline model

Figure 3: Baseline IRF to Monetary Policy Shock: Industrial Commodities



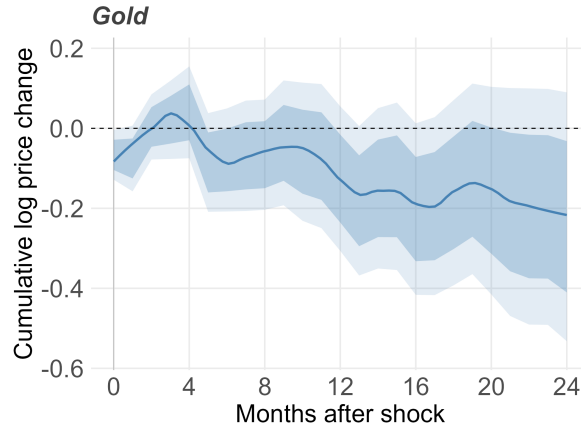
Note: Cumulative log price response to a one-unit exogenous increase in the 1-year US Treasury yield, instrumented by [Acosta et al. \(2025\)](#). Dark and light shaded areas represent 68% and 90% confidence intervals respectively, based on Newey-West HAC standard errors. Sample: 1994–2026.

Figure 4: Baseline IRF to Monetary Policy Shock: Agricultural Commodities



Note: See [Figure 3](#).

Figure 5: Baseline IRF to Monetary Policy Shock: Gold



Note: See [Figure 3](#). Gold is included as a reference commodity; see [Section 4](#) for a discussion of the interpretation of its results.

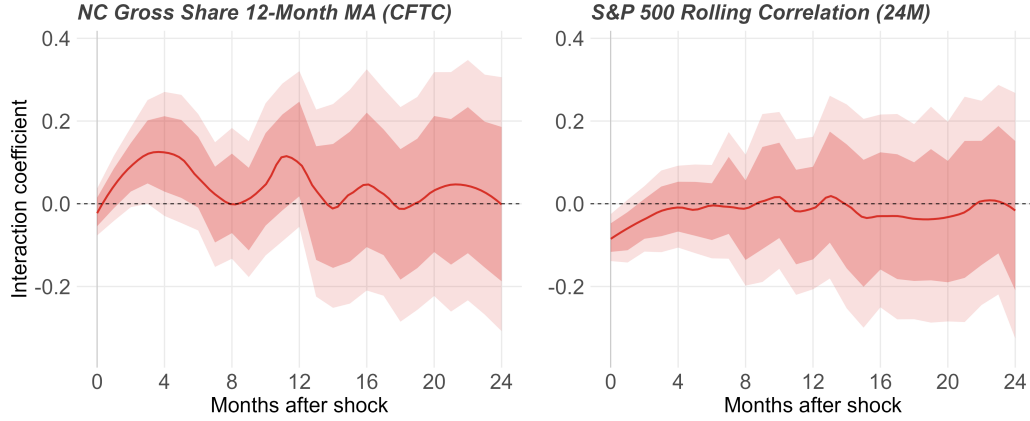
The baseline IRFs reveal some heterogeneity in the price response to monetary policy shocks across commodities. Gold and soybeans display a negative and statistically significant cumulative response, with the 68% confidence interval excluding zero from approximately month 12 and 10 onward respectively and the effect persisting through the full 24-month horizon. Note that gold also shows a significantly negative result for the immediate impact, with a subsequent rebound and then the definitive decline.

Oil shows no statistically significant average response at any horizon, with the IRF oscillating around zero throughout. Copper displays a positive response over the medium horizon, significant at the 68% level between months 4 and 16, before reverting towards zero at longer horizons. A weak positive trend with wide confidence intervals is found for wheat, yielding no significant result at the 90% level.

These null or even positive patterns motivate the interaction model, which allows the price response to vary with the degree of financialization rather than imposing a constant average effect. The intuition is that probably the low-financialization period is averaging out the effect of the highly financialized one, at least for highly and consistently financialized commodities such as copper and oil.

4.2 Interaction model

Figure 6: Interaction Effect of Financialization on Monetary Policy Transmission - Gold

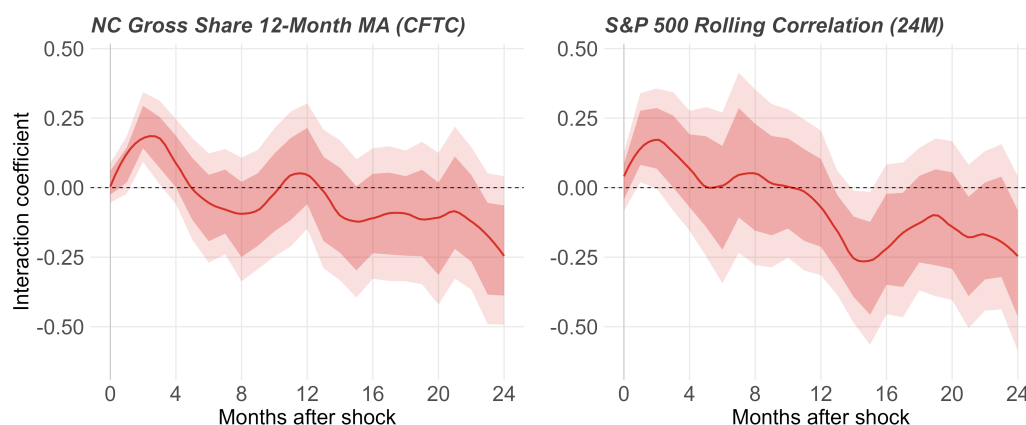


Note: Interaction coefficient $\hat{\phi}^h$ from equation (6), estimated separately for each financialization measure. A negative coefficient indicates that higher financialization amplifies the negative price response to monetary tightening. The financialization measure is standardised within commodity using full-sample moments. Endogenous variable: change in 1-year US Treasury yield, instrumented by [Acosta et al. \(2025\)](#). Dark and light shaded areas represent 68% and 90% confidence intervals respectively, based on Newey-West HAC standard errors with bandwidth $\max(1, h)$. Sample: 1994–2026.

The interaction model shows no significant pattern in results for gold when monetary policy is interacted with financialization measured through correlation with S&P500 index returns, which is consistent with the evidence of this measure oscillating between negative and positive values around zero.

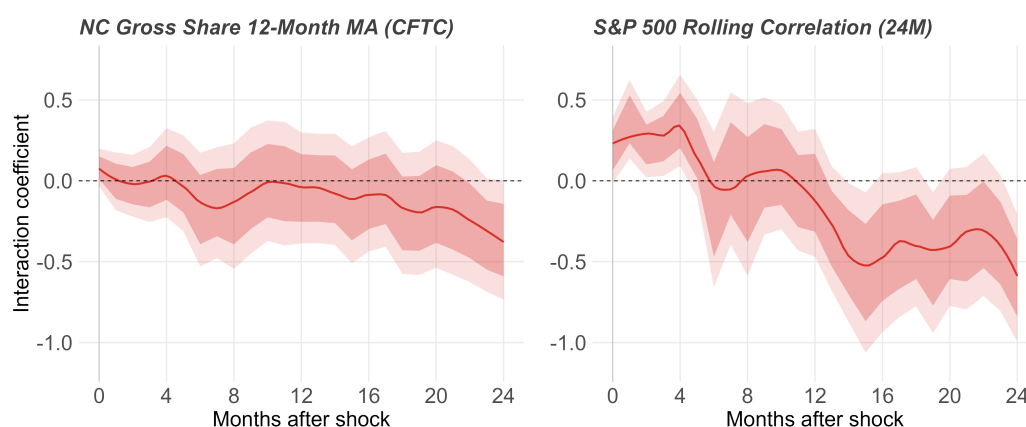
The NC gross share interaction exhibits a positive point estimate over the short horizon — the opposite sign from the risk appetite channel prediction, which is negative, persistent and medium-term as we show below. In fact, gold is highly financialized, but probably different types of financialization coexist for gold.

Figure 7: Interaction Effect of Financialization on Monetary Policy Transmission: Copper



Note: See [Figure 6](#).

Figure 8: Interaction Effect of Financialization on Monetary Policy Transmission: Oil

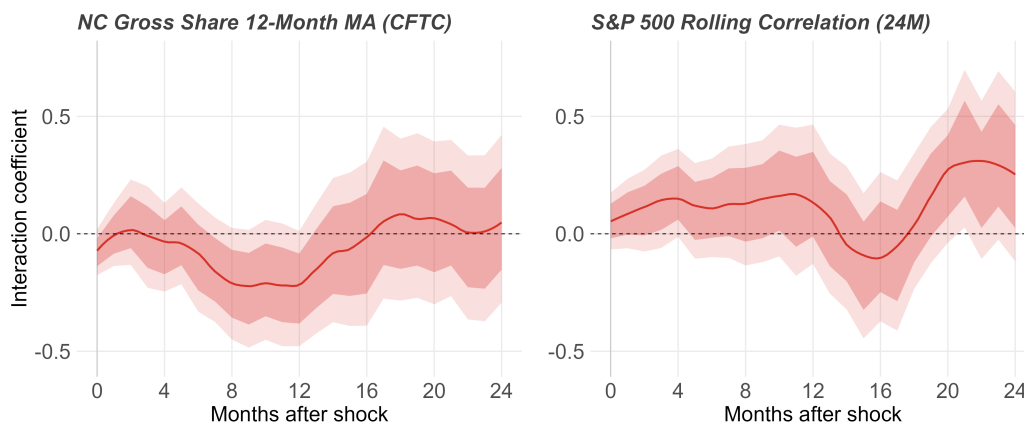


Note: See [Figure 6](#). Oil additionally includes contemporaneous and four lags of the ? OPEC supply news shock as a control.

Industrial commodities - copper and oil - are the strongest results. As it can be seen from [Figure 7](#) and [Figure 8](#), when interacted with the intensive margin measure for financialization, a monetary policy tightening has a negatively amplified effect significantly different from zero at 68% confidence level from month 12 onwards. Notice that both have an initially significant positive bump, and a subsequent decline. These trends are consistent with the highly financialized environment of industrial commodities.

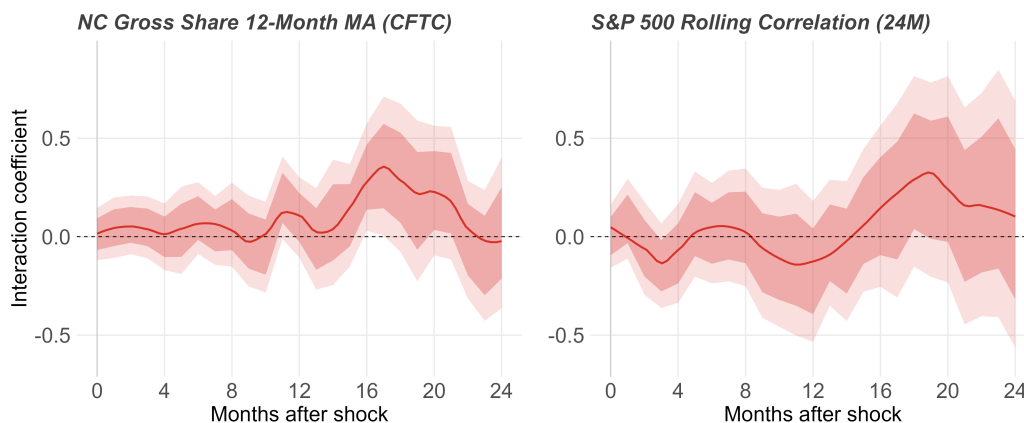
The extensive margin, the NC gross share measure, shows instead no significant pattern for both copper and oil across all horizons. The IRF is flat for oil, while it is flat with an initial positive bump for copper, consistent with the S&P500 correlation's results.

Figure 9: Interaction Effect of Financialization on Monetary Policy Transmission: soybeans



Note: See [Figure 6](#).

Figure 10: Interaction Effect of Financialization on Monetary Policy Transmission: Wheat



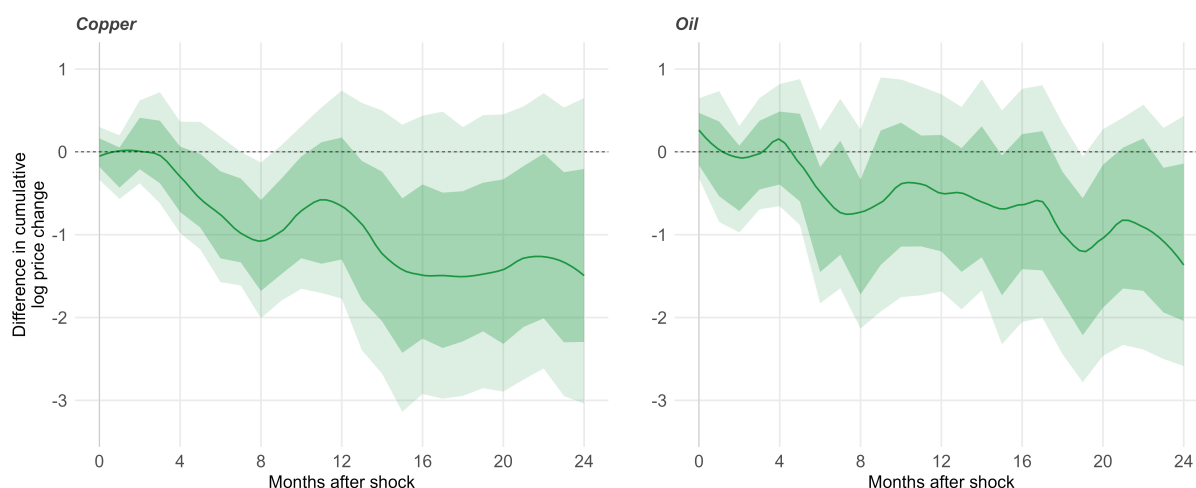
Note: See [Figure 6](#).

Agricultural commodities show non-significant results throughout the horizons except for some case which seem to be more casual than causal - e.g., the negative coefficients at 68% for lags between 8 and 12 of the NC gross share for soybeans. The shape of the IRFs is highly irregular, with

quite wide confidence bands. No clear pattern emerges, consistently with the idea that agricultural commodities were less financialized than industrial commodities. In [Section A.4](#) I run the same interaction model for agricultural commodities excluding de-financialization years - post 2015 - but results show only limited change in terms of significance.

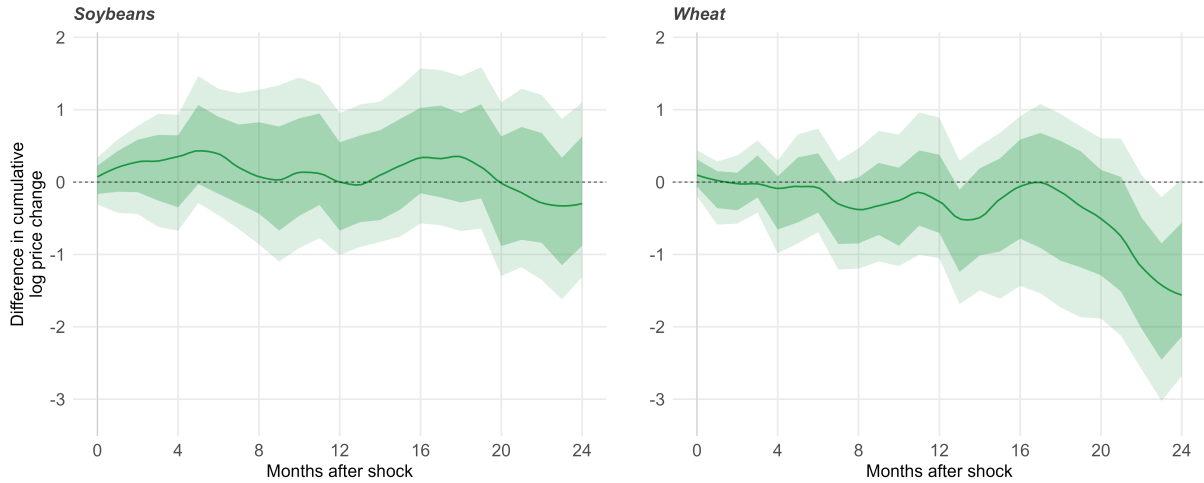
4.3 Subsample comparison

Figure 11: Change in Monetary Policy Transmission - Industrial Commodities



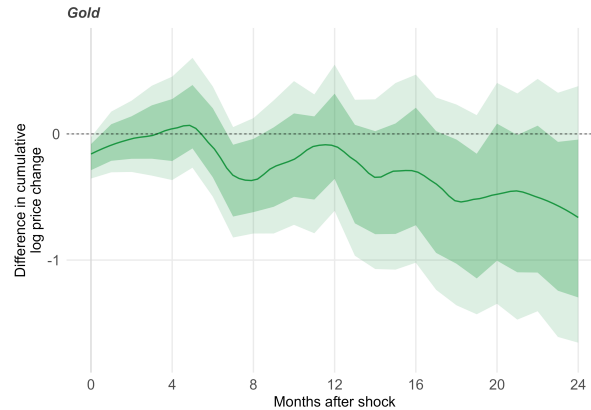
Note: Difference in cumulative log price response between the post-financialization subsample (2010–2026) and the pre-financialization subsample (1994–2007), $\Delta \hat{\delta}^h = \hat{\delta}_{\text{post}}^h - \hat{\delta}_{\text{pre}}^h$. A negative value indicates stronger negative price response post-financialization. Confidence bands from a time-cluster block bootstrap with block length 12 months and $B = 500$ replications; dark and light shaded areas represent the 16th–84th and 5th–95th percentiles respectively. The years 2008–2009 are excluded from both subsamples.

Figure 12: Change in Monetary Policy Transmission - Agricultural Commodities



Note: See [Figure 11](#).

Figure 13: Change in Monetary Policy Transmission - Gold



Note: See [Figure 11](#).

The post-financialization period shows consistently more negative results for monetary policy tightening for copper, oil and gold - copper remaining significant even at 90% confidence level. Soybeans and wheat show almost no change and display very wide confidence bands.

A notable feature of the subsample results is that the negative post-minus-pre difference for copper, oil, and wheat does not arise from a near-zero pre-financialization response becoming negative post-financialization, but rather from a positive pre-financialization IRF reversing sign - or being

annulled - in the post-financialization period. In the pre-financialization subsample, monetary tightening is associated with rising prices for these commodities. In the post-financialization period this relationship reverses, with the risk appetite channel generating a negative price response as financial investors reduce their commodity futures exposure following a contractionary shock. The full-sample baseline IRF is consequently flat or positive for these commodities — not because monetary policy has no effect, but because two opposing regimes of approximately equal magnitude cancel in the time-series average. Detailed pre and post IRFs are reported separately in [Section A.6](#).

4.4 Discussion

In short, three main findings emerge. First, monetary policy impact analysis on these commodity futures gives inconclusive results over the whole period when using a classic LP-IV model, with the exception of gold. Secondly, introducing the interaction term and the cross-commodity comparison shows that financialization amplified the effects of monetary policy in highly-financialized industrial commodities - oil, copper -, but not in agricultural commodities, despite a significant rise in the presence of non-commercial traders in these markets; results for gold were inconclusive. Heterogeneity is also present across financialization measures, with the intensive margin - S&P500 correlation - delivering the above results as opposed to the extensive margin - NC gross share. Finally, the monetary policy impact did change between the pre and post financialization periods and it changed in the expected direction at least for the same commodities - copper, oil, gold; however, a significant portion of this difference is due to a counterintuitive effect of monetary policy in the pre-financialization period - i.e., a price increase with monetary policy tightening.

The first finding is not particularly surprising: positive and significant effects before financialization and - mildly - negative and significant after explain the null overall effect even better than null before and negative after, as previously hypothesized.

The analysis of the interaction between financialization and monetary policy confirms that where

financialization was more intense, monetary policy effects were amplified and more in line with the behaviour of a conventional financial asset. Moreover, the use of two different measures offers an interesting insight about this phenomenon. This research unveils that the rising number of non-commercial traders operating on commodity futures markets does not seem to cause any difference in monetary policy transmission *per se*. Rather, this non-commercial inflow seems to be a necessary but not sufficient condition. If this inflow is present and brings co-movement with equity returns, then one observes the classic negative impact of monetary policy tightening on commodity prices we were expecting due to financialization.

Interestingly, this pattern emerges in the medium run, which means other forces act on these prices immediately after the impulse. Furthermore, the growth in correlation with the equity index is noisier and weaker than the growth in non-commercial traders presence, which suggests a lagged and less than one-to-one correspondence. Finally, this preeminence of the intensive over the extensive margin is confirmed by the comparison of industrial and agricultural commodities, whose differences in financialization patterns emerge more strikingly in the S&P 500 measure⁹. This partial divergence could be explained by a different kind of financialization motives across different commodity classes: while financialization in industrial commodities is indeed driven by speculation and risk appetite, it may be that for agricultural commodities the portfolio-diversification strategy is still predominant - i.e., we are dealing with an index-like type of investment. If this is the case, it is logical that monetary policy regime changes do not necessarily make investors push prices in one direction or the other.

A closer look at the interaction model's results adds a meaningful piece to the puzzle allowing me to identify three channels operate simultaneously with different signs and time profiles from the big picture emerged.

At short horizons, the storage cost channel seems to dominate. The storage cost theory is an historically well known principal driver of commodity futures prices ([Fama and French, 1987](#); [Frankel and Rose, 2009](#); [Casassus and Collin-Dufresne, 2005](#)). This theory explains futures price move-

⁹See [Figure 1](#) and [Figure 2](#)

ments through the cost of carrying physical inventory, which is closely related to the interest rate. As a matter of fact, higher rates - contractionary monetary policy - generate immediate upward pressure on commodity prices: higher interest rates raise the cost of carrying physical inventory, requiring futures prices to adjust upward to reflect this cost movement.

This channel operates independently of financialization and is present across subsamples, explaining part of the positive pre-financialization IRF and the initial positive bump in the interaction coefficient at short horizons. When futures markets are highly financialized, the risk appetite channel accumulates at medium horizons and the storage cost effect fades due to arbitrage: financial investors complete their portfolio rebalancing away from commodity futures, generating a negative and persistent price response whose magnitude increases with the degree of financialization intensity.

In the pre-financialization period, when financial investor participation was limited, the fundamental channels dominated and the net IRF was positive. Post-financialization, the risk appetite channel overwhelmed fundamental channels, reversing the sign of the IRF. The full-sample baseline IRF is consequently flat for oil, copper, and wheat not because monetary policy is ineffective, but because two opposing regimes of approximately equal magnitude cancel in the time-series average.

However, as previously noted, the storage channel can only partially explain why in the pre-financialization era, the IRF are consistently positive for copper and oil. In particular, [Casassus and Collin-Dufresne \(2005\)](#) find that the convenience yield there is strong mean reversion and spot price anchorage for these commodities futures, which rules out a long-term positive effect. In any case, disentangling which forces lead to this is beyond the scope of this analysis, given that regardless of this aspect, the sub-sample comparison does show IRFs that move in the right direction in the high financialization period, becoming negative for copper and null - but declining - for oil.

Gold warrants a separate discussion. As discussed in [Section 2.2](#), both measures are not very informative for gold because multiple forces act on gold prices.

Nevertheless, the baseline on full sample and in the sub-samples comparison tell an interesting story. Gold is overall mildly negatively reacting to monetary policy tightening and the difference between the two periods does seem to be significantly negative - thus implying a shift in a direction coherent with our expectations. No positive bump is present coherently with the absence of the storage channel drawn from [Casassus and Collin-Dufresne \(2005\)](#) results - which can be considered evidence its presence in the other commodities.

Then, the absence of a clear trend in the intensive margin - together with the spuriousness of the extensive margin measure previously illustrated - may point in the direction of a contradiction with our analysis. Nonetheless, this is not the case: as a non-yielding asset whose opportunity cost equals the real risk-free rate directly, gold prices respond negatively to monetary tightening as investors substitute towards interest-bearing assets. The post-financialization period may have slightly amplified this channel through a scale effect: if this measure captured a real trend, the doubling of the NC gross share from approximately 15% to 30% expanded the base of rate-sensitive financial investors responding to the real interest rate signal, generating a proportionally larger aggregate price response to the same monetary shock. Thus, gold response may be differently motivated, but still somewhat related to this financialization phenomenon.

5 Conclusions

This paper examined the effects of commodity futures financialization on monetary policy transmission, documenting heterogeneity across commodity categories and isolating the element of the financialization process that appears to trigger the change, i.e., the co-movement with equity prices.

Taken together, the three models deliver a consistent directional message: the financialization of commodity markets strengthened monetary policy transmission, with the risk appetite channel generating a negative price response to monetary policy tightening that was absent or dominated by fundamental channels in the pre-financialization period. Indeed, even where significance is not achieved, the results remain directionally consistent with the hypothesis, and the absence of signif-

icance should be attributed at least in part to limited statistical power rather than to the absence of an underlying effect. These results advance the discussion on commodity futures financialization, which had focused mostly on the extensive margin and on its effect on commodity price levels rather than monetary policy transmission.

Future research could build on this work in several directions. First, the extensive margin financialization measure could be improved by drawing on the CFTC Disaggregated Commitments of Traders report, available from 2006, which decomposes non-commercial positions into swap dealers and managed money traders. This would allow to clean the extensive measure from index-trading. The limited historical availability of this series motivated the use of the legacy COT report here, but constructing a comparable long-run series would be valuable.

Second, the heterogeneity of impact across commodities could be enlarged by extending the analysis to other financialized futures markets, including non-US benchmark contracts such as LME copper or ICE Brent crude. Given that US futures prices serve as global benchmarks through arbitrage, consistent results in non-US markets would substantially strengthen the external validity of the findings.

Third, future work could model the role of financialization within a structural VAR framework, which should deliver more precise impulse response functions and richer channel identification.

Finally, more attention should be devoted to the positive price response to monetary policy tightening in the pre-financialization period, tentatively attributed here to the commodity supercycle but deserving more rigorous identification.

Overall, addressing these questions would enrich policymakers' understanding of how monetary policy decisions propagate through commodity markets, contributing to the understanding of how monetary policy propagates through financial markets into commodity prices.

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A Appendix

A.1 Commodity futures selection

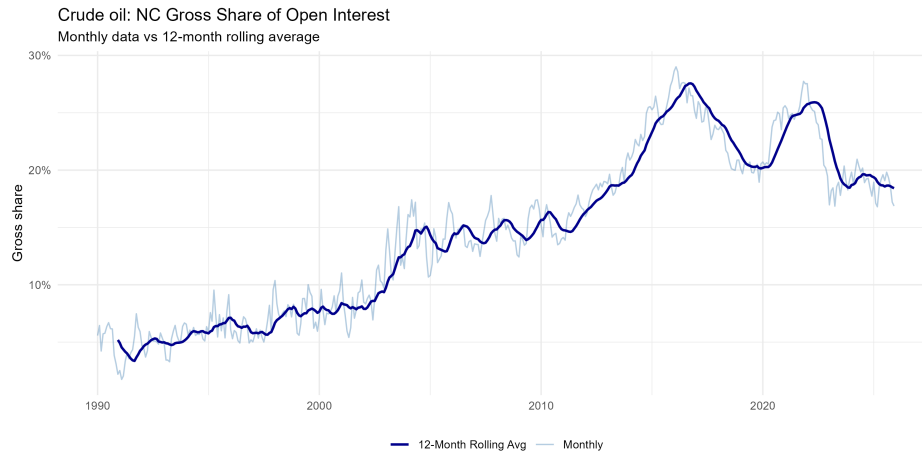
Table 1: Commodity Futures Contracts

Commodity	Benchmark	Exchange	Unit
Crude oil	WTI front-month	NYMEX/CME	USD per barrel
Gold	COMEX front-month	COMEX/CME	USD per troy ounce
Copper	COMEX front-month	COMEX/CME	USD per pound
soybeans	CBOT front-month	CME Group	USD per bushel
Wheat	CBOT front-month	CME Group	USD per bushel
Coffee	ICE Arabica front-month (Coffee C)	ICE US	USD per pound

Note: All series are front-month futures contracts retrieved from Refinitiv (LSEG) Datastream. Monthly returns are computed as log differences of end-of-month settlement prices.

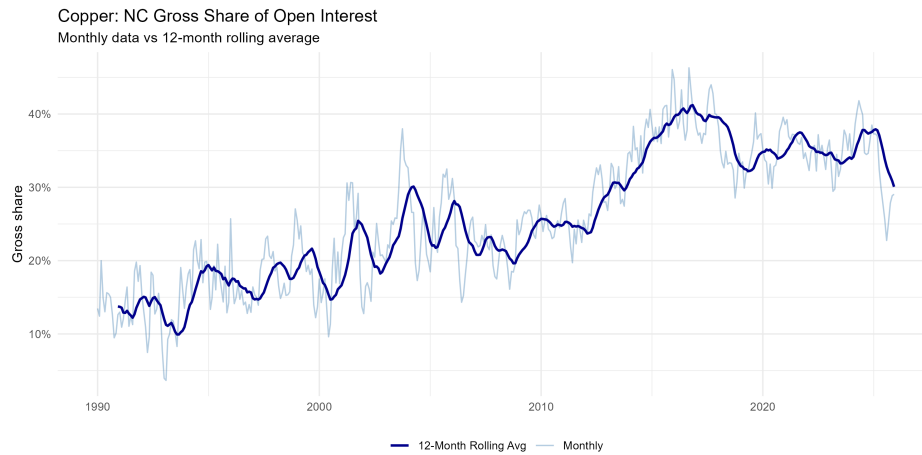
A.2 Financialization measures - single commodity and index breakdown

Figure A.1: NC Gross Share of Open Interest: Crude Oil



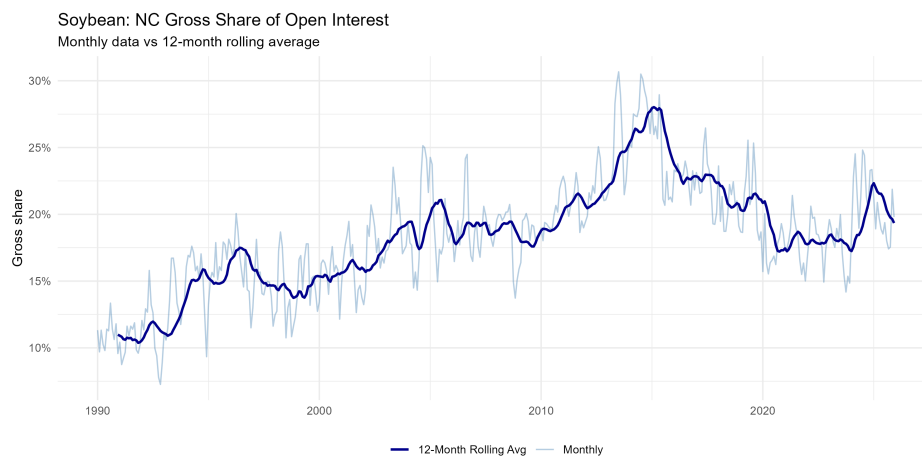
Note: Monthly non-commercial gross share of open interest for crude oil futures, computed from CFTC Commitments of Traders data. The dark line represents the 12-month rolling average; the light line represents the monthly series. Sample: 1990–2026.

Figure A.2: NC Gross Share of Open Interest: Copper



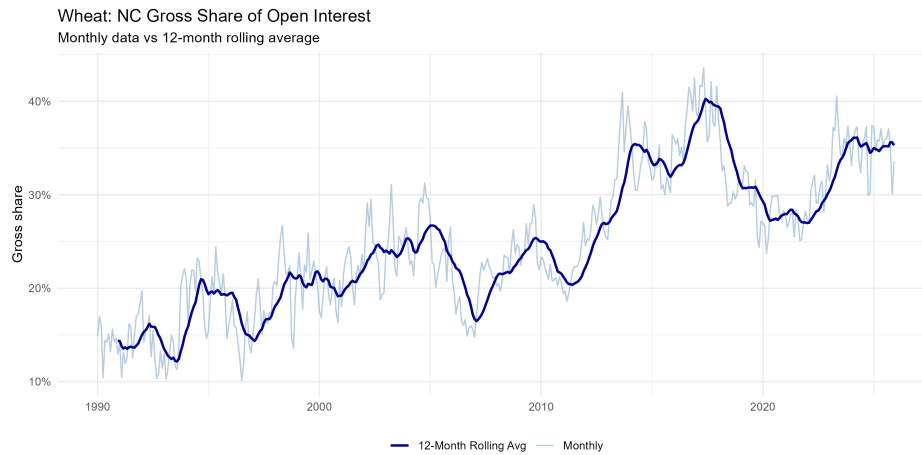
Note: Monthly non-commercial gross share of open interest for copper futures, computed from CFTC Commitments of Traders data. The dark line represents the 12-month rolling average; the light line represents the monthly series. Sample: 1990–2026.

Figure A.3: NC Gross Share of Open Interest: Soybean



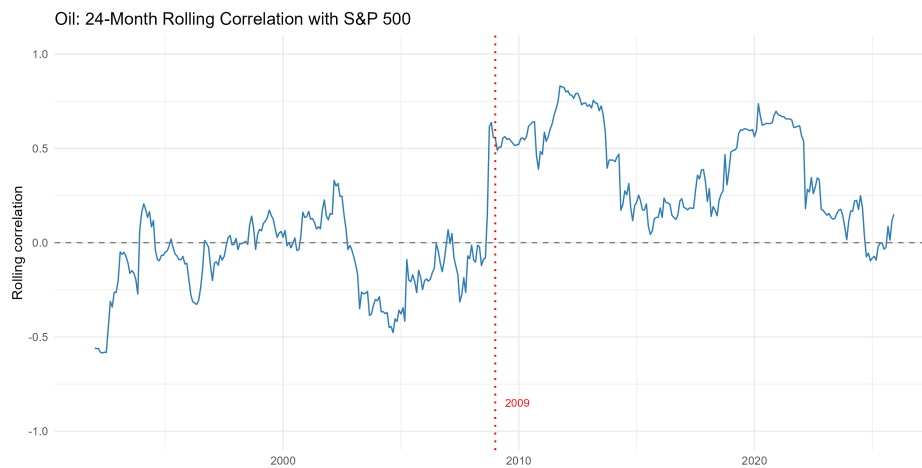
Note: Monthly non-commercial gross share of open interest for soybean futures, computed from CFTC Commitments of Traders data. The dark line represents the 12-month rolling average; the light line represents the monthly series. Sample: 1990–2026.

Figure A.4: NC Gross Share of Open Interest: Wheat



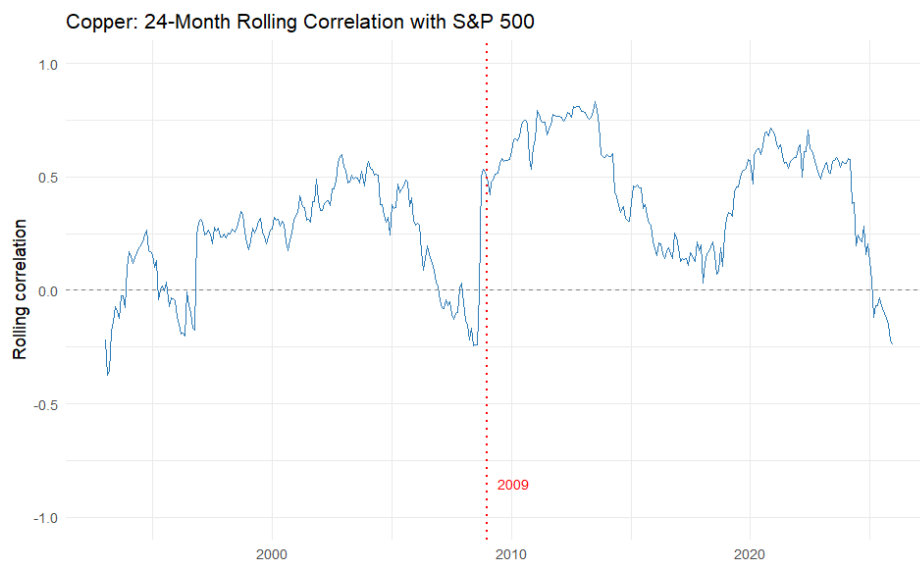
Note: Monthly non-commercial gross share of open interest for wheat futures, computed from CFTC Commitments of Traders data. The dark line represents the 12-month rolling average; the light line represents the monthly series. Sample: 1990–2026.

Figure A.5: 24-Month Rolling Correlation with S&P 500: Crude Oil



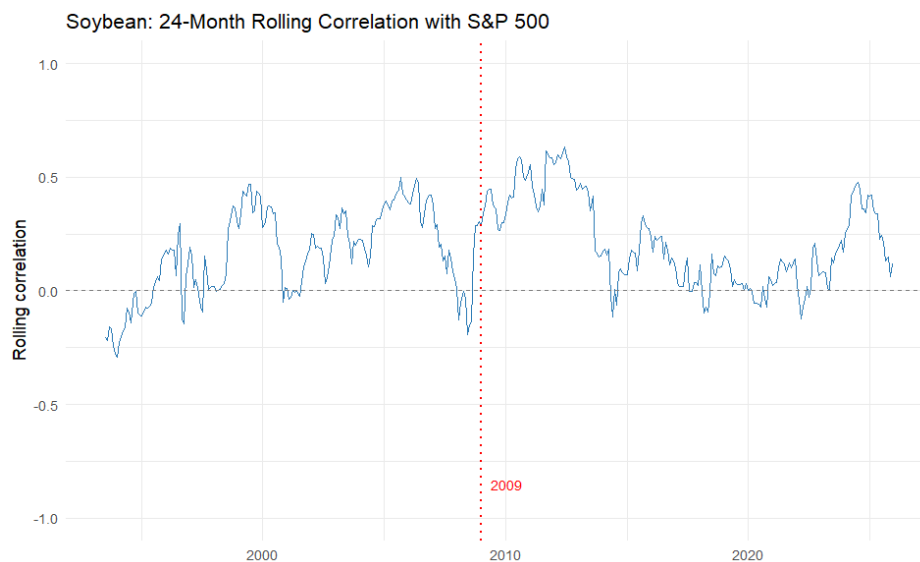
Note: 24-month rolling correlation between monthly log returns of crude oil futures and the S&P 500 index. The red dotted vertical line marks January 2009, associated with the onset of large-scale financialization following the Global Financial Crisis. Sample: 1990–2026.

Figure A.6: 24-Month Rolling Correlation with S&P 500: Copper



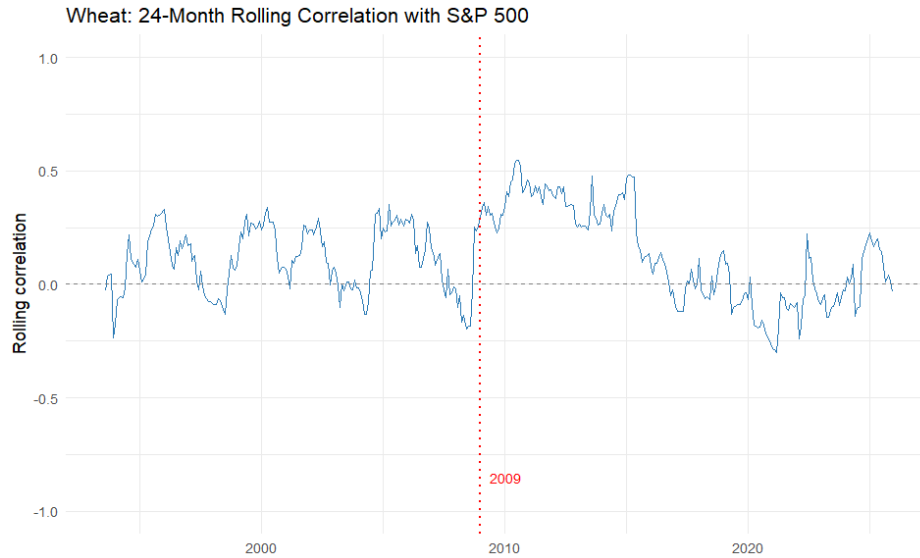
Note: 24-month rolling correlation between monthly log returns of copper futures and the S&P 500 index. The red dotted vertical line marks January 2009, associated with the onset of large-scale financialization following the Global Financial Crisis. Sample: 1990–2026.

Figure A.7: 24-Month Rolling Correlation with S&P 500: Soybean



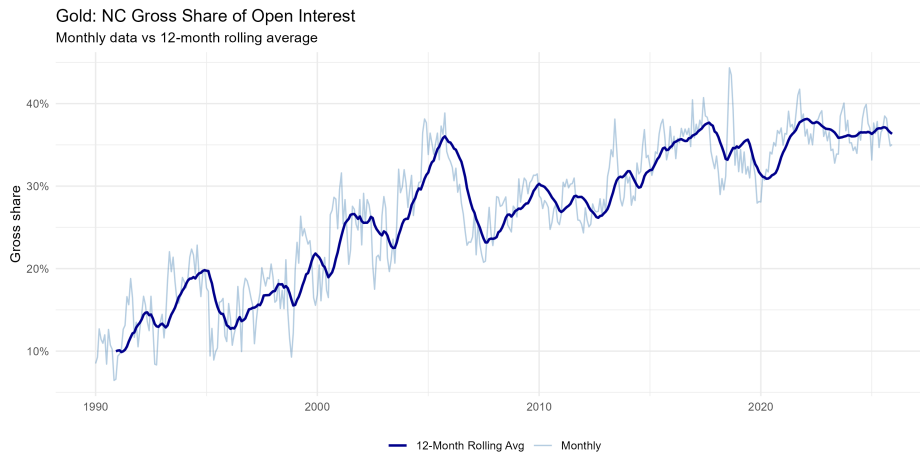
Note: 24-month rolling correlation between monthly log returns of soybean futures and the S&P 500 index. The red dotted vertical line marks January 2009, associated with the onset of large-scale financialization following the Global Financial Crisis. Sample: 1990–2026.

Figure A.8: 24-Month Rolling Correlation with S&P 500: Wheat



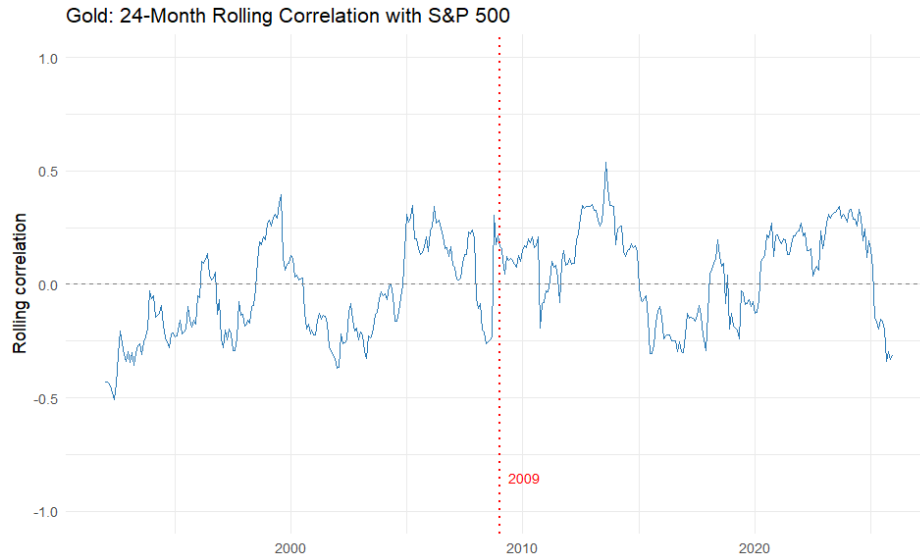
Note: 24-month rolling correlation between monthly log returns of wheat futures and the S&P 500 index. The red dotted vertical line marks January 2009, associated with the onset of large-scale financialization following the Global Financial Crisis. Sample: 1990–2026.

Figure A.9: NC Gross Share of Open Interest: Gold



Note: Monthly non-commercial gross share of open interest for gold futures, computed from CFTC Commitments of Traders data. The dark line represents the 12-month rolling average; the light line represents the monthly series. Sample: 1990–2026. The NC gross share for gold reflects the classification of bullion banks as commercial traders under the legacy CFTC COT framework and does not reliably measure financial investor participation; it is therefore not used as a financialization measure in the empirical analysis.

Figure A.10: 24-Month Rolling Correlation with S&P 500: Gold



Note: 24-month rolling correlation between monthly log returns of gold futures and the S&P 500 index. The red dotted vertical line marks January 2009, associated with the onset of large-scale financialization following the Global Financial Crisis. Sample: 1990–2026. Gold’s dual role as a risk asset during expansions and a safe haven during crises produces an unstable correlation with equity returns that does not reliably capture financial integration; this measure is therefore not used in the empirical analysis for gold.

A.3 Checks

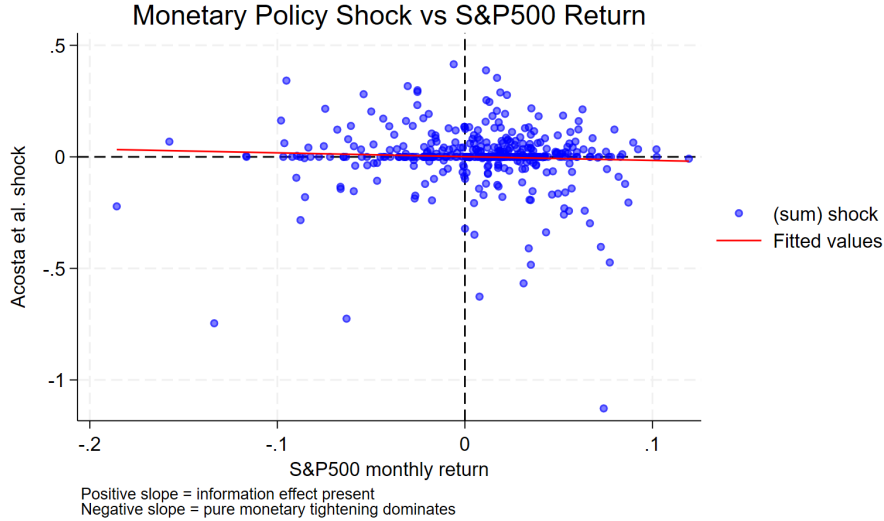
This section reports two diagnostic checks for the validity of the instrumental variable strategy. [Table 2](#) presents the first stage results confirming instrument relevance. [Figure A.11](#) tests for the information effect by examining the correlation between the monetary policy shock and contemporaneous equity returns.

Table 2: First Stage Results

	Δi_t
Acosta et al. shock	0.480*** (0.075)
Constant	−0.000 (0.011)
Observations	383
F-statistic	41.40
R^2	0.098

Note: OLS regression of the monthly change in the one-year Treasury constant maturity yield Δi_t on the Acosta et al. (2025) high-frequency monetary policy surprise. Standard errors in parentheses. The F-statistic comfortably exceeds the conventional threshold of 10 for instrument relevance. Sample: 1994–2026. *** $p < 0.01$.

Figure A.11: Monetary Policy Shock and S&P 500 Returns



Note: Scatter plot of the monthly Acosta et al. (2025) monetary policy shock against contemporaneous S&P 500 log returns. The red dashed line represents the fitted values from a simple OLS regression. The absence of a significant positive correlation rules out a systematic information effect whereby FOMC announcements convey positive news about economic conditions; the slightly negative slope is consistent with pure monetary tightening, whereby contractionary surprises reduce equity valuations through the risk appetite channel. Sample: 1994–2026.

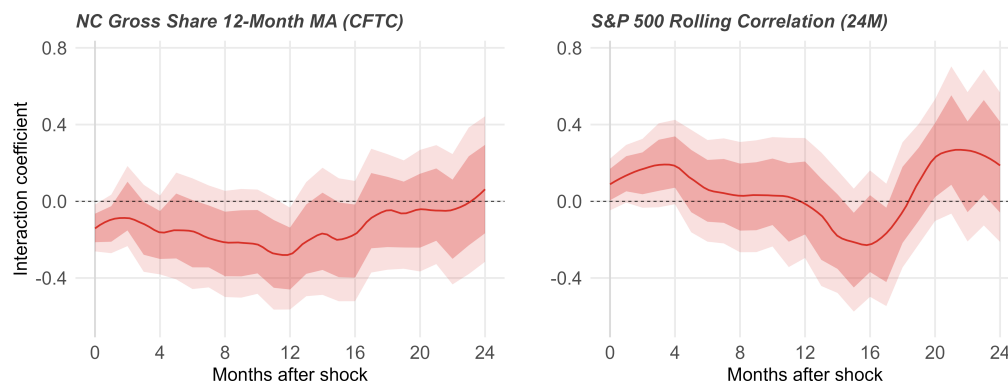
A.4 Capped Sample for Agricultural Commodities

For soybeans and wheat, the interaction model is additionally estimated on a sample capped at December 2017. The motivation is the following: the post-2015 period for agricultural commodities is characterised by a quite significant de-financialization pattern, as shown in [Section 2.2](#): both measures decline substantially after 2015 for all three commodities, suggesting that the structural relationship between financialization and monetary policy transmission may have changed. Capping at December 2017 excludes the de-financialization period at the cost of a shorter time series. The idea is the following: if we do not consider de-financialization, we should find more significant results even for agricultural commodities, despite weaker financialization.

The capped specification is identical to equation (6), with the restriction that only observations up to December 2017 enter the regression. Financialization is standardized using full-sample moments throughout, ensuring that the interaction coefficients remain comparable with the full-sample estimates. Crucially, standardization is always performed using full-sample moments, even when the regression is restricted to a sub-period, ensuring comparability of interaction coefficients across specifications.

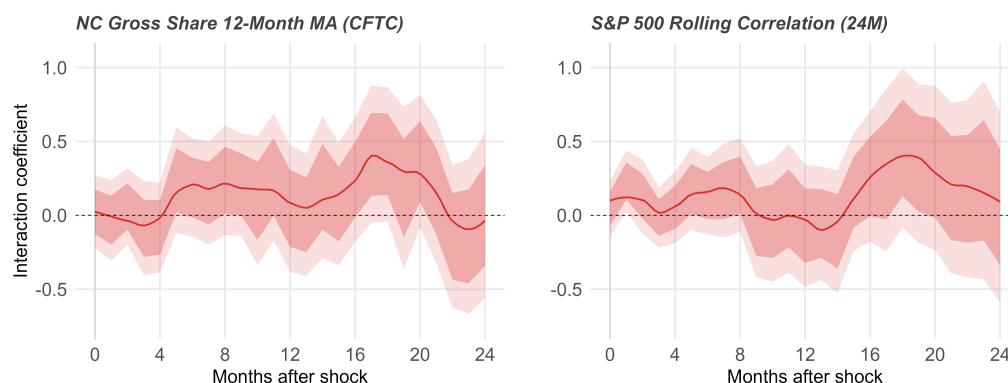
However, results improve only mildly. For soybeans, the interaction coefficient under the NC gross share measure turns negative and marginally significant between months 6 and 18, consistent with a moderate amplification effect during the peak financialization period. For wheat, however, the S&P 500 correlation interaction remains insignificantly different from zero or positive across all horizons, and the NC gross share interaction delivers mixed results. Overall, the capped sample experiment provides partial support for the financialization hypothesis for soybeans but results remain mixed and lack the consistency observed for industrial commodities.

Figure A.12: Interaction Effect of Financialization on Monetary Policy Transmission: soybeans (Capped Sample 1994–2017)



Note: Interaction model with sample restricted to 1994–2017 to exclude the post-2015 de-financialization period in agricultural commodity markets.

Figure A.13: Interaction Effect of Financialization on Monetary Policy Transmission: Wheat (Capped Sample 1994–2017)



Note: Interaction model with sample restricted to 1994–2017 to exclude the post-2015 de-financialization period in agricultural commodity markets..

A.5 Subsample comparison specification details

This appendix justifies the regressor reduction and describes the time-cluster block bootstrap procedure used to construct confidence bands for the difference in impulse responses between the pre- and post-financialization subsamples.

The parsimonious model. The regressor reduction in the subsample comparison is necessary given the degrees of freedom constraints imposed by splitting the sample into two non-overlapping

subsamples. With a pre-financialization subsample of approximately 168 monthly observations (1994–2007) and a post-financialization subsample of approximately 180 observations (2010–2026), the observations-to-parameters ratio at long horizons becomes critically low under the full control set. At horizon $h = 24$, the baseline specification with currency controls and four crisis dummy lags would require estimating approximately 30 parameters on fewer than 140 effective observations per subsample after accounting for lag construction, leaving insufficient degrees of freedom for reliable inference. The parsimonious specification decreases the number of regressors to 20 parameters while preserving the core identification structure of four lags of the price change, industrial production growth, and inflation, plus contemporaneous and two lags of the crisis dummies.

Bootstrap rationale. Comparing whether the separate confidence bands for $\hat{\delta}_{\text{pre}}^h$ and $\hat{\delta}_{\text{post}}^h$ overlap is not a valid test for a structural change in monetary policy transmission. The variance of the difference satisfies:

$$\text{Var}(\hat{\Delta}\delta^h) = \text{Var}(\hat{\delta}_{\text{post}}^h) + \text{Var}(\hat{\delta}_{\text{pre}}^h) - 2 \text{Cov}(\hat{\delta}_{\text{post}}^h, \hat{\delta}_{\text{pre}}^h) \quad (8)$$

Because both impulse responses are estimated from the same underlying commodity futures price series, the covariance between $\hat{\delta}_{\text{post}}^h$ and $\hat{\delta}_{\text{pre}}^h$ is likely positive and substantial. This reduces $\text{Var}(\hat{\Delta}\delta^h)$, overlapping confidence bands are not an insightful criterion for statistic significance.

Bootstrap Procedure. Confidence bands for $\hat{\Delta}\delta^h$ are constructed using a time-cluster block bootstrap following [Berthold et al. \(2026\)](#). Let $\mathcal{T}_{\text{pre}}$ and $\mathcal{T}_{\text{post}}$ denote the sets of dates in the pre- and post-financialization subsamples respectively. For each replication $b = 1, \dots, B$:

1. Draw blocks of 12 consecutive months with replacement separately from $\mathcal{T}_{\text{pre}}$ and $\mathcal{T}_{\text{post}}$, re-

taining all commodity observations within each selected month. Resampling separately for each subsample preserves the within-subsample time-series structure while allowing the two periods to vary independently across replications.

2. Retain the first-stage fitted values $\widehat{\Delta}_{i_t}$ fixed within each replication, since resampling occurs only at the second stage. This correctly separates first-stage uncertainty, which is treated as negligible given the full time-series sample, from second-stage sampling uncertainty.
3. Re-estimate the second stage separately for each commodity i and each subsample, and compute:

$$\widehat{\Delta}\delta^{h,*(b)} = \hat{\delta}_{\text{post}}^{h,*(b)} - \hat{\delta}_{\text{pre}}^{h,*(b)}, \quad h = 0, \dots, H \quad (9)$$

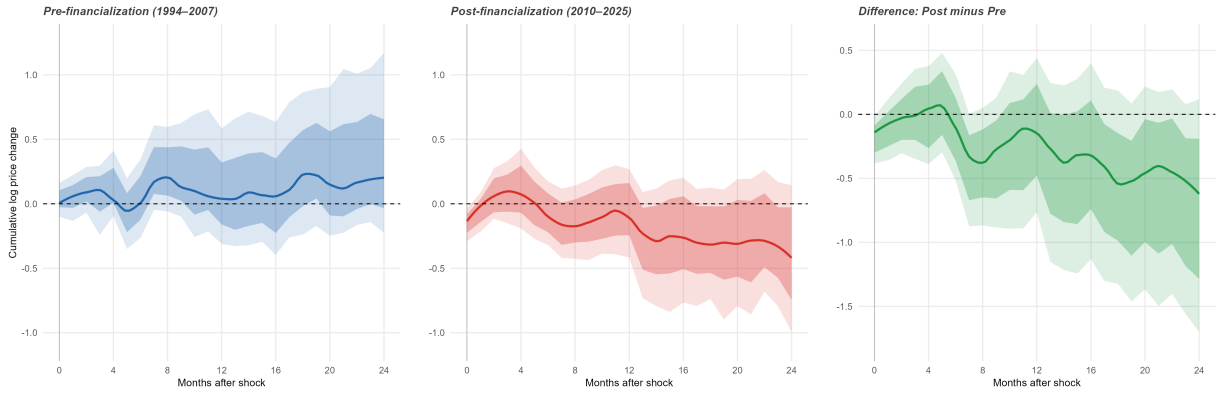
Concretely, the pre-period dates $\mathcal{T}_{\text{pre}}$ are divided into blocks of 12 consecutive months, and the bootstrap draws these blocks randomly with replacement to construct a synthetic time series of the same total length. The same is done independently for $\mathcal{T}_{\text{post}}$. Within each block, the chronological order is perfectly preserved — the correct serial dependence, the correct relationship between the dependent variable and the regressors, and the correct lag structure are all identical to the original data. The only departure from the original series occurs at the joins between blocks, where the transition from one block to the next is artificial. The block length of 12 months is chosen to preserve a full annual cycle of serial dependence — capturing the main autocorrelation structure of monthly commodity price data — while still allowing sufficient variability across replications. Because the second stage regression estimates a relationship between the monetary policy shock and the cumulative price response that is assumed stable within each subsample, what matters is having a representative sample of shock-response pairs from each period, which the block bootstrap provides while preserving the local serial dependence that affects standard error estimation.

Confidence bands are the 16th–84th and 5th–95th percentiles of $\{\widehat{\Delta}\delta^{h,*(b)}\}_{b=1}^B$, corresponding to 68% and 90% coverage respectively. By re-estimating both subsample regressions jointly within each bootstrap draw, the procedure implicitly captures the covariance between $\hat{\delta}_{\text{post}}^h$ and $\hat{\delta}_{\text{pre}}^h$ without

requiring analytic estimation. We set $B = 500$ replications and $H = 24$ months. Confidence bands for the individual pre- and post-period impulse responses are constructed analogously from the marginal distributions of $\hat{\delta}_{\text{pre}}^{h,*(b)}$ and $\hat{\delta}_{\text{post}}^{h,*(b)}$.

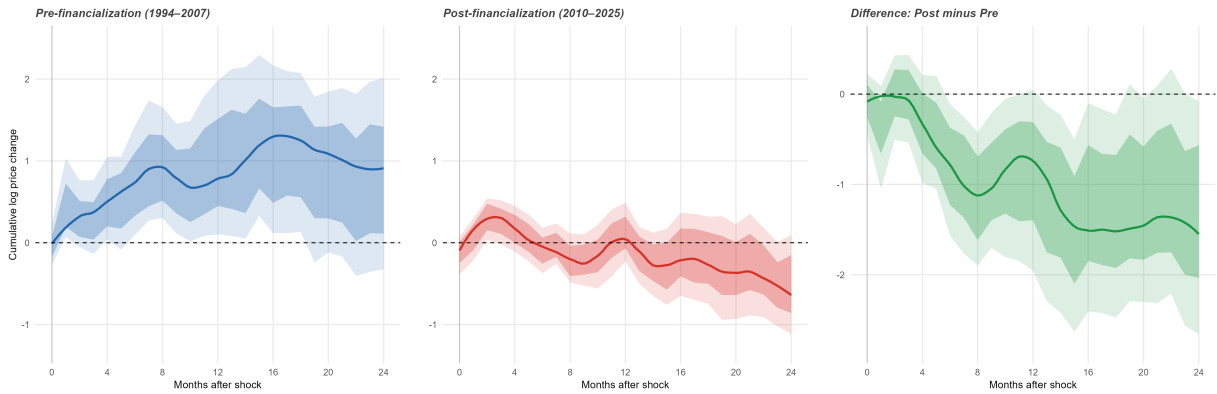
A.6 Bootstrap Result breakdown for Subsample Comparison

Figure A.14: LP-IV IRF Pre vs Post Financialization: Gold



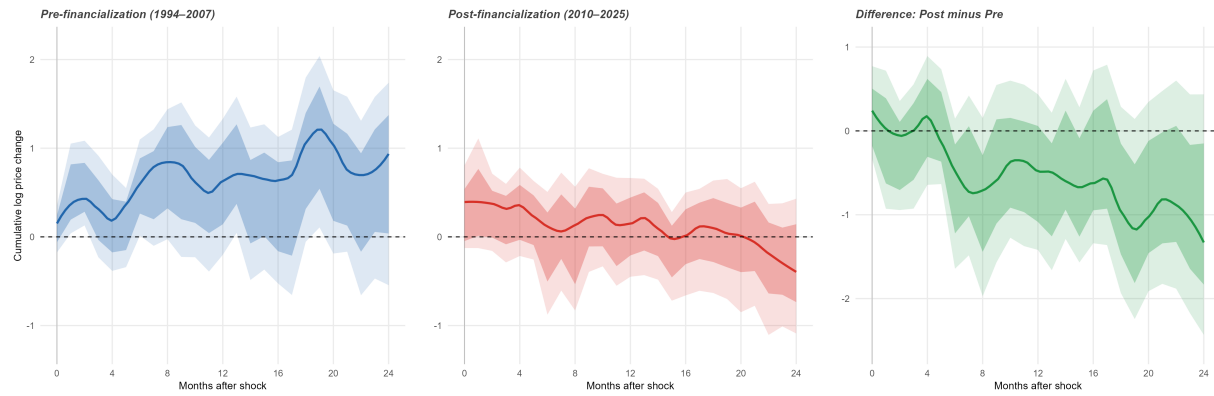
Note: Cumulative log price response to a one-unit exogenous increase in the 1-year US Treasury yield, estimated separately on the pre-financialization subsample (1994–2007, blue) and the post-financialization subsample (2010–2026, red). The right panel reports the difference $\Delta\hat{\delta}^h = \hat{\delta}_{\text{post}}^h - \hat{\delta}_{\text{pre}}^h$ (green). The first stage is estimated separately for each subsample; the second stage uses a parsimonious control set comprising four lags of the log price change, industrial production growth, and inflation, plus crisis dummies with two lags. Confidence bands are constructed from a time-cluster block bootstrap with block length 12 months and $B = 500$ replications; dark and light shaded areas represent the 16th–84th and 5th–95th percentiles of the bootstrap distribution respectively. The years 2008–2009 are excluded from both subsamples.

Figure A.15: LP-IV IRF Pre vs Post Financialization: Copper



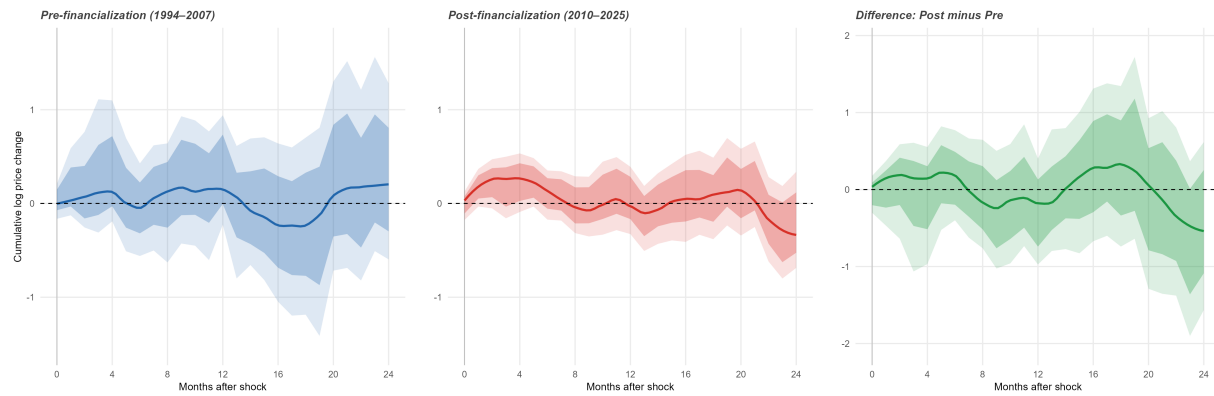
Note: See [Figure A.14](#).

Figure A.16: LP-IV IRF Pre vs Post Financialization: Oil



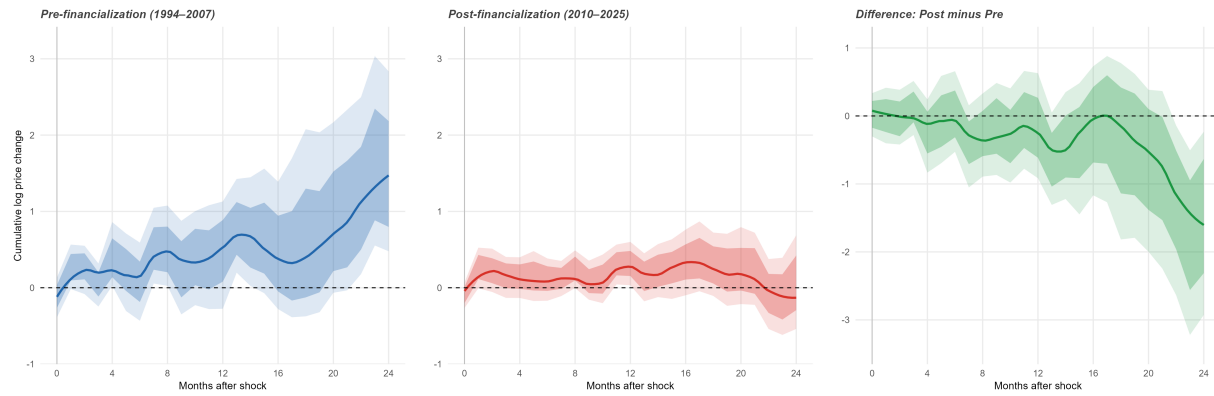
Note: See [Figure A.14](#).

Figure A.17: LP-IV IRF Pre vs Post Financialization: soybeans



Note: See [Figure A.14](#).

Figure A.18: LP-IV IRF Pre vs Post Financialization: Wheat



Note: See [Figure A.14](#).